

Markov State Models: a methodological study on the HP35 villin headpiece

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I. INTRODUCTION

A. Brief description of the biological problem, knowns and unknowns

Markov State Models (MSMs) have been used for nearly two decades to extract kinetic information from molecular dynamics simulations and provide a coarse-grained description of the conformational space of biomolecules [6]. The classical pipeline for building MSMs consists of several steps, each involving a number of choices that can affect the quality of the final model. Although procedures have standardized over the years, still the process consists of stages where more than one choice is possible and free parameters must be optimized in lack of a clear minimization objective, making the process appear somewhat "handcrafted". Recently, deep learning approaches have been proposed to automate some parts or all of the process of building MSMs - most importantly, feature selection and dimensionality reduction stages [8][9], with the aim is improving the quality of the resulting models and reducing the human-induced bias.

B. Defining the question

In this work we review the classical pipeline of MSM construction, concentrating on the feature selection and dimensionality reduction stages. The aim is to review the classical approaches before moving on to the deep learning ones, and setting up a baseline for model comparison that can be extended to comprehend the deep learning methods in future studies.

C. Defining the plan: translating the question on the language of the computer (brief)

We use the HP35 villin headpiece as a case study, a small protein that has been extensively studied in the literature and for which long molecular dynamics simulations are available. We compare two sets of features: contact distances and dihedrals. We compare two dimensionality reduction methods: PCA and TICA. We compare the resulting models in terms of accuracy of the folding time estimate, ability to provide a structural characterization of the metastable states, and robustness to hyperparameter choice.

II. METHODS

A. Dataset assembly

All analyses presented in this work are based on a single 300 μ s molecular dynamics trajectory of the fast-folding Lys24-Nle/Lys29-Nle mutant of the 35-residue villin headpiece subdomain (HP35), simulated at 360 K. HP35 is a prototypical ultrafast-folding protein with an experimentally measured folding time of approximately 0.7 μ s at this temperature and has become a standard benchmark system for Markov state model construction and validation [3–5]. The trajectory was originally generated by D. E. Shaw Research on the Anton supercomputer and reported by Piana *et al.* [1]. Their simulations predicted a folding time of approximately 1.5 μ s, reasonably close to the experimental estimate of 0.73 μ s for the same mutant, although still larger by a factor of about $\times 2.5$. At the time of this work, the original trajectory generated by D. E. Shaw Research was temporarily unavailable due to maintenance of the data repository. We therefore used a preprocessed version distributed by Nagel *et al.* through their GitHub repository, which has been adopted in recent benchmark studies of Markov state modeling [3, 4].

B. Methods used for analysis (in case you developed a method, it goes into Results)

All code is written in Python and is available on GitHub at . Markov state models were built using the Deeptime library [7]. Since MSM construction is a multi-step process, it is especially important to understand clearly the choices made at each step and validate them before moving on to the next one. The principle "garbage in, garbage out" applies here, and a mistake at an early stage can propagate and affect the final results. For this reason, in this section we describe synthetically the motivation for each step and define the methods that were investigated and compared in this work.

1. Feature selection: contact distances and dihedrals.
2. Dimensionality reduction: PCA and TICA.
3. Clustering into microstates.
4. MSM estimation: selection of lag time. Validation: implied timescales test.

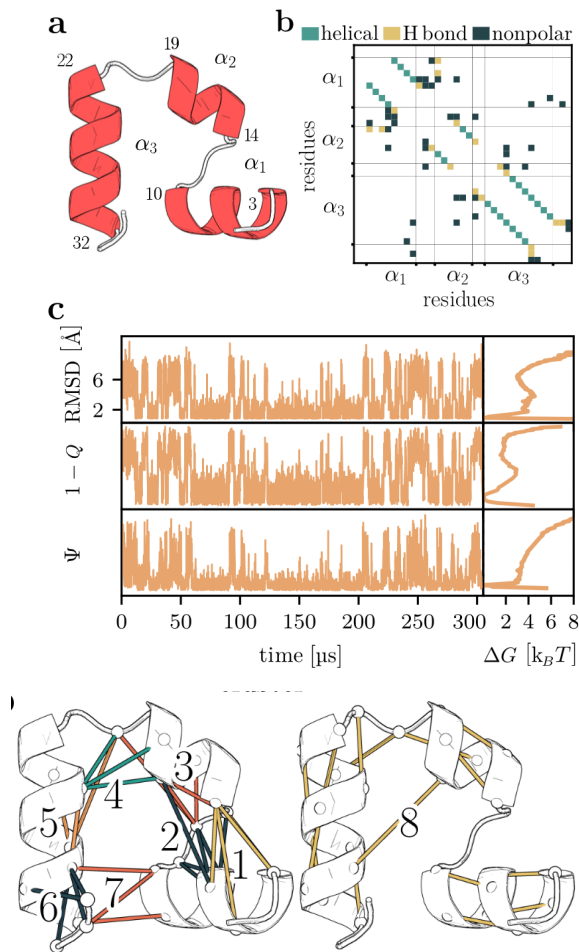


FIG. 1: *Folding of villin headpiece (HP35).* (a) Molecular structure, consisting of three α -helices (residues 3 – 10, 14 – 19 and 22 – 32) that are connected by two loops. (b) Contact map showing the 42 native contacts of HP35. (c) Time evolution of the RMSD, compared to $1 - Q$ (with Q being the fraction of native contacts) and Ψ representing the sum over the backbone dihedral angles ϕ_i of the three α -helices. The right side shows the corresponding free energy curves (in units of $k_B T$) along these coordinates.

5. Coarse-graining into macrostates: PCCA and MPP. Validation: CK test.

Feature selection

This is the first step. If all degrees of freedom were included, the system would trivially have a markovian dynamics. But this would not be useful, since the model would not be able to provide any insight on the system. The aim of feature selection is to select a subset of degrees of freedom that are relevant for the process under study, and that can provide a good description of the dynamics. In this work we compare two sets of features: contact distances and dihedrals.

Specifically, we analyzed the following time series of different internal coordinate representations:

- hp35.mindists2: all atom pairwise distances occurring more frequently than 30% [nm];
- hp35.crystaldists: the atom distances of all contacts occurring in the crystal structure 2f4k [2] [nm];
- hp35.dihs: backbone dihedral angles [degrees].

Refer to the paper by Nagel *et al.* [3] for details on the feature selection.

Dimensionality reduction

The aim is to compress the input coordinates (features) into a lower-dimensional space that still carries *kinetic* information. The notion of geometric distance in this space should ideally reflect the kinetic distance, meaning that two points that are close in this space should have a high probability to interconvert in a short time, while two points that are far should have a low probability to interconvert in a short time. Indeed, the next stage is clustering the samples into a finite set of microstates, and the clustering is based on the geometric distance in the reduced space. If the reduced space does not reflect the kinetic distance, then the clustering will not be able to capture the metastable states of the system, and the resulting MSM will not be able to provide a good description of the dynamics.

Any dimensionality reduction will distort the free energy landscape by some extent. The goal is to keep the distortion as low as possible.

For this reason, the dimensionality reduction stage is a crucial one. However, there is no clear objective function to optimize, and the choice of the method and the number of dimensions to retain is often based on heuristics and trial-and-error. Actually, recently a variational approach has been proposed. In this work we set a baseline, with the comparison two methods: PCA and TICA. PCA is a linear method that finds the directions of maximum variance in the data, while TICA is a linear method that finds the directions of maximum autocorrelation in the data. TICA is often considered more suitable for MSM construction, since it is designed to capture the slow dynamics of the system, which are often the most relevant for the process under study. However, PCA is still widely used, and it is interesting to compare the As a baseline, we will check if we find indeed that the tICA model is better as we expect.

C. Statistical analysis

Here talk about Bayesian uncertainty estimation, implied timescales test, CK test. Maybe show tables and

figures of the results of the statistical analysis here? Something, the other can go in SI. Also estimation of the folding time.

III. RESULTS

This is the real talk, the biological talk! Comparative analysis of the results of the various MSM constructed.

Sections describing different aspects of your analysis (3a-3d). All sections should be organised as follows, the text is continuous, broken only to a few paragraphs in each section:

- Question studied in the section and the goal of this calculation.
- Few sentences on how it was done, mostly referring to datasets and approaches in the Methods.
- Actual results and observations, each supported by either a i) table or ii) figure or iii) number in the text.
- You need to interpret the observation. For example, do not only write $X \not\sim Y$, but also the meaning of this relationship with respect to your question.
- When presenting the results, refer to the respective Figure and Table, where it was shown.
- You can add more sentences for the interpretation. Please use precise phrases for the level of evidence, if the results 'suggest', 'indicate', 'show', 'demonstrate', 'support', 'corroborate'. Do not put too much 'hypothesis', and 'propose' these are synthetic phrases.
- End each section with a conclusion sentence for example, 'Taken together the results consistently show that X is larger than Y, indicating that X was an ancestor of Y'.

IV. DISCUSSION

Here you try to put your results into a broader context and present your scientific view/model based on that. This is a continuous text, broken into paragraphs. In case you have two distinct 'huge' conclusions, you may generate separate sections (max 2). Chapter 4 should be organised as follows:

- You may phrase your biological question again.
- Briefly summarise the results (max one paragraph). You do not repeat the words you used earlier, make it concise. Make it evident how your results answer the question you have asked.

- Compare your results with something in the literature (use PubMed, or literature indicated in UniProt, PDB or any datasets we learnt about). 1-2 articles enough. The point is that you
- show how your results relate to previous ones. Different scenarios: 'support', 'in contrast', 'in addition', 'gives insights into'.
- If possible, present a model. This can be a molecular 'mechanism', 'scenario', 'reason'. For example: 'A and B belong to the same family'; 'mutation XD likely changes the interaction in liver'; 'in disease T we have less phosphorylation sites or the level of phosphorylation decreased'; 'in this experiment, there is a problem with degradation of protein P.' etc.
- The model can be presented as: 'propose', 'suggest', 'indicate', 'hypothesize', or even 'support' or 'demonstrate'. If possible try to present a cartoon figure or scheme.

V. CONCLUSIONS

One paragraph, emphasizing the main finding and formulate a message for the future (perspective), how to continue from here, what kind of further studies are needed. You may have Conclusion as chapter 5.

VI. REFERENCES

The text should contain references, to previous results, data, methods, softwares, etc.. Format is unrestricted, but you should use these refs in the text, and list them at the end. I expect ca 15-20 refs listed.

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 - [3] D. Nagel, S. Sartore, and G. Stock, *Selecting Features for Markov Modeling: A Case Study on HP35*, Journal of Chemical Theory and Computation 2023 19 (11), 3391-3405 DOI: 10.1021/acs.jctc.3c00240.
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- [8] A. Mardt, L. Pasquali, H. Wu and F. Noé, *VAMPnets for deep learning of molecular kinetics*, Nature Communications 9(1):1-11, 2018.
- [9] C. Wehmeyer and F. Noé. *Time-lagged autoencoders: deep learning of slow collective variables for molecular kinetics*. J. Chem. Phys., 148(24):241703, 2018. DOI: 10.1063/1.5011399.

SUPPORTING INFORMATION